

HUMAIRA ALI

Early-career Digital Specialist Engineer pursuing a Bachelor of Technology in Electronics and Communication Engineering

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering student with hands-on experience in embedded systems, data analysis, AI/ML-enabled applications, and product development. Experienced in building ESP32-based intelligent systems, analyzing sensor data, developing automation solutions, and collaborating on customer-focused products. Passionate about Product Analytics, AI-driven decision making, and building technology solutions that create measurable business impact.

SKILLS

Programming :	python, SQL, Verilog, C, C++, Java, HTML, MySQL
Data & Analytics :	Data Analysis, Data Visualization, Microsoft Excel, Power BI, Pandas, NumPy, SQLite, MySQL, Tableau
Soft Skills :	Communication, Leadership, Teamwork, Problem Solving, Critical Thinking, Adaptability, Time Management, Collaboration, Decision Making
Tools & Platforms :	MATLAB, Keil uVision, LTSpice, Proteus, SQLite, GitHub, Git, Linux, Power BI, Excel, VS Code, Jupiter Notebook, Tableau, Docker, GitHub Actions
Product & Business :	Product Analysis, Root Cause Analysis, Product Lifecycle, Stakeholder Communication, Data Visualization, SQL Based Analytics, Relational Data Modeling, Work Flow Automation, Data Cleaning
Languages :	English, Hindi, Arabic, Urdu

PROJECTS

Smart Helmet System for Accident Detection & Navigation

- Designed and developed an ESP32-based intelligent safety system to address the high rate of two-wheeler accidents in India by integrating multi-sensor data, real-time accident detection, Bluetooth communication, and GPS-enabled emergency alerts. Analyzed sensor data and optimized detection thresholds to improve accuracy while minimizing false emergency notifications.
- Designed and developed an **ESP32-based smart helmet** for real-time accident detection and emergency response to improve rider safety.
- Integrated **MPU6050 and ultrasonic sensors** to enable crash detection, rear-obstacle awareness, and rider assistance using a multi-sensor embedded system.
- Implemented a **3G threshold-based accident detection algorithm** with a **10-second user cancellation window**, Bluetooth communication, and GPS-enabled SOS alerts to minimize false emergency notifications.
- Leveraged **AI/ML research** to analyze accident patterns, evaluate detection techniques, and optimize the system for improved reliability and real-world performance. Included **Analog Layout Concepts** and **microcontrollers** in the design process.

Technologies / Tools Used : ESP32, Embedded C++, Bluetooth, MPU6050, Ultrasonic Sensor, Android, Artificial Intelligence, Machine Learning, microcontrollers

RFID-Based Multi-Module Campus Management System

- Developed an **ESP32- and RFID-based campus management platform** integrating attendance, library management, and cafeteria billing across 3 operational modules.
- Automated attendance logging, PDF bill generation, Excel exports, and database management through a **Python GUI application**.

Technologies / Tools Used : ESP32, Python, SQLite, RFID, Bluetooth, Tkinter

Visa Prediction MLOps Pipeline

Built an end-to-end MLOps pipeline to predict US visa application approval using automated data ingestion, validation, feature engineering, model training, evaluation, and deployment.

Developed production-ready ML workflows with MongoDB Atlas, MLflow experiment tracking, SMOTEENN for class imbalance, and FastAPI for real-time browser and REST API predictions.

Containerized the application with Docker and implemented GitHub Actions CI/CD for automated testing, model training, and deployment workflows.

Technologies / Tools Used : Python, Fast API, Scikit-learn, MongoDB, ML-flow, Docker, GitHub-Actions

WORK EXPERIENCES

Intern | Internship

Bharat Heavy Electricals Limited (BHEL)

May 2026 - Jun 2026

Bhopal, Madhya Pradesh

Developed PLC-based industrial automation solutions to improve operational efficiency. Collaborated with engineers to analyze manufacturing workflows, troubleshoot control systems, and support embedded firmware development. Contributed to process optimization through automation and real-time monitoring.

Technologies / Skills Used : Semiconductor Fundamentals, Embedded Firmware Development

CERTIFICATIONS

- Introduction to Programming in C** — IIT Kharagpur
- Internet of Things** — IIT Kharagpur
- MATLAB Signal Processing** — MathWorks
- VLSI Design & Verification Program** — Ethnus
- Google IT Support Professional Certificate** — Google
- Introduction to Self-Driving Cars** — University of Toronto
- Bits and Bytes of Computer Networking** — Google

RELEVANT COURSEWORK

- Data Structures & Algorithms
- Database Management Systems
- Operating Systems
- Computer Networks
- Machine Learning & Artificial Intelligence
- Embedded Systems
- Digital Electronics
- Microprocessors
- Software Engineering
- VLSI
- Signals and Systems

EDUCATIONS

Vellore Institute Of Technology

Bachelor of Technology in Electronics and Communication Engineering

2023 - 2027

CGPA : 8.66 / 10

- 10th: **89.8/100**, Carmel Convent BHEL-CBSE, 2021
- 12th: **80.4/100**, Carmel Convent BHEL-CBSE, 2023